

NIPUN BALACHANDRAN NAIR

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PROFESSIONAL SUMMARY

Dedicated AI Engineer and Researcher with expertise in developing scalable machine learning systems and fine-tuning Large Language Models (LLMs). Currently pursuing MPhil at Monash University, focusing on Generative AI and User Simulation. Proven track record of delivering innovative solutions across automotive, finance, and linguistic AI sectors. Strong foundation in Python, deep learning frameworks, and cloud technologies with hands-on experience in building production-ready AI applications.

EDUCATION

Master of Philosophy (M.Phil) - Faculty of Information Technology, Department of Data Science & AI

Monash University, Clayton, Melbourne, Australia | 2025 - 2027 (Supervisors: Dr Teresa Wang, Dr Tongtong Wu)

- Research Focus: Generative AI, User Simulation, Textual gradient optimization
- Currently working on Automatic optimization of Agentic workflows and prompts in user behavior simulation

Bachelor of Technology (B.Tech) - Computer Science

Amrita Vishwa Vidyapeetham, Bangalore, India | 2020 - 2024 (CGPA: 8.48/10)

- Final year Project on Classification of stages of “Serpiginous Choroiditis” images from Narayana Hospital. Compared performance of deep learning models, VGG16, RESNET50, DenseNet, MobileNet and found RESNET 50 with the highest performance in recall, accuracy, precision, and AUC score.

PUBLICATIONS

- “Joint Workflow and Prompt Optimization for User Behavior Simulation” under review for TOIS 2026 Journal
- “Prompt Optimization for User Simulation in Recommender Systems: A Multi-Objective Framework” KDeLLM Workshop, ICDE Conference, 2026 | [[Paper Link](#)]
- "Building a LLaMA2-Fine-Tuned LLM for Odia Language Utilizing Domain Knowledge Instruction Sets" AI-ML Systems Conference (ACM Published), 2024 | [[Paper Link](#)]
- "Breast Cancer Classification using Machine Learning Algorithms" IEEE Medical Imaging Conference, 2022 | [[Paper Link](#)]

EMPLOYMENT HISTORY

Industry Intern & Graduate Engineer Trainee | Jan 2024 - Dec 2024

Stellantis | Bengaluru, India

- Developed a Service and Test Application Macchina API framework for ICAN, ISPAAK and IDM Persistence App Framework
- Demonstrated Cybersecurity concepts of Discrete Access Control, Mandatory Access control, User and Group Permissions, SystemD and POSIX capabilities in a presentation to senior management
- Assisted in the development of automotive prototypes, contributing to timely project milestones and enhancing product innovation

Industry Intern | Jul 2023 – Oct 2023

OdiaGenAI | Remote

- Trained and fine-tuned LLaMA-2 for low-resource Odia NLP, achieving strong generation and summarization performance (ROUGE-L: 0.65, BLEU: 0.61).
- Built and evaluated multilingual (Odia & Hindi) datasets and models, delivered a live technical demo, and collaborated with a distributed research team to translate findings into improvements in performance of models.

Research Intern | Jun 2022 - Aug 2022

UC Santa Cruz, California | Santa Cruz, California, USA

- Analyzed global bank data using Python, enhancing data accessibility for research projects
- Extracted stock data from EDGAR, improving data accuracy for the Global Capital Allocation Project
- Mentored Ph.D. students in Python debugging, fostering collaborative learning and skill development

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, Java, Scala, MATLAB, SQL, C, C++

AI/ML Frameworks: TensorFlow, PyTorch, scikit-learn, Hugging Face Transformers, LangChain

LLM Technologies: OpenAI GPT-4, Claude, LLAMA2, Fine-tuning, Prompt Engineering, RAG Systems

Cloud & DevOps: AWS (EC2, S3, Bedrock), Docker, Git, CI/CD Pipelines

Data Engineering: Apache Spark, Hadoop, Data Preprocessing, Feature Engineering

Specializations: NLP, Computer Vision, Conversational AI, Agentic Systems, Time-Series Analysis

KEY PROJECTS & RESEARCH

Agentic AI Systems Development

Personal Research Project | 2026 - Present

- Developed autonomous AI agents using OpenAI GPT models with memory capabilities and tool integration
- Built email automation agent with natural language processing for intelligent response generation
- Implemented conversational loops with context preservation and multi-turn dialogue management
- Technologies: Python, OpenAI API, LangChain, Agent Orchestration Frameworks

LLAMA2 Fine-tuning for Low-Resource Languages

Academic Research | 2023 - 2024

- Fine-tuned LLAMA2 model on Odia domain knowledge dataset, achieving significant performance improvements
- Developed 'Olive' - instruction-following Odia GPT model for enhanced human-AI interactions
- Created and curated specialized datasets for low-resource language model training
- Technologies: PyTorch, Hugging Face, Parameter-Efficient Fine-Tuning (PEFT), LoRA

Breast Cancer Classification System

IEEE Published Research | 2023

- Developed machine learning pipeline for breast cancer classification achieving 95%+ accuracy
- Implemented ensemble methods combining multiple classification algorithms for robust predictions
- Published findings in IEEE conference proceedings, contributing to medical AI research
- Technologies: Python, Scikit-learn, Pandas, Numpy, Medical Imaging Processing

Computer Vision for Heritage Preservation

Academic Research Project | 2023

- Implemented Fully Convolutional Networks (FCN) for Malayalam palm leaf manuscript segmentation. Applied deep learning techniques for cultural heritage digitization and preservation.
- Technologies: TensorFlow, OpenCV, Image Segmentation, Deep Learning

Environmental Data Analysis using Kalman Filtering

Data Science Project | 2022

- Analyzed Bangalore pollution data (2018-2021) using Kalman filtering for state estimation
- Predicted air quality trends across multiple pollutants (PM2.5, PM10, NO2, SO2, CO, O3) gasses.
- Technologies: Python, Time-Series Analysis, Statistical Modelling, Data Visualization

TEACHING & MENTORSHIP

Teaching Assistant - FIT1059: AI for Everyone (under Dr TT Wong)

Monash University, Melbourne | Semester 2, 2025, 2026

- Delivered engaging tutorials on foundational AI principles and practical implementations, supporting diverse student cohorts and improving conceptual understanding.
- Recognized for effective teaching and student engagement through positive Student Evaluation feedback.

CERTIFICATIONS:

- GEN AI PINNACLE PROGRAM | Nov 2023 - Jun 2025 | *Analytics Vidhya*
- GOOGLE CLOUD SKILLS | Jun 2023| *Google*
- AWS CERTIFICATION BADGE | May 2023 - Jun 2023| *Amazon*
- SKILL NATION CERTIFICATION FOR BUSINESS INTELLIGENCE USING MICROSOFT POWER BI | May 2023|*Microsoft*
- Data Structures and Algorithms FOR JAVA | Jun 2023 - Dec 2023| *Apna College*
- MACHINE LEARNING FOR SOIL AND CROP MANAGEMENT IIT KHARAGPUR NPTEL | Jan 2022 - Apr 2022| *IIT Kharagpur*
- TAKE CHARGE OF YOUR TIME | Jul 2022| *Ankur Warikoo*
- ICON OF AMRITA PRESTIGIOUS FRESHMEN ANNUAL AWARD | Oct 2020| *Amrita Vishwa Vidyapeetham*

INTERESTS:

Languages: English (Native), Hindi (Fluent), Malayalam (Fluent)

Research Interests: Generative AI, Large Language Models, Small Language Models, Conversational AI, Retrieval-Augmented Generation (RAG), AI Agents, Multilingual NLP, AI Ethics & Responsible AI, User Simulation

Technical Interests: MLOps & Model Deployment, Distributed Computing, Real-time AI Systems, Open Source Contributions, Cloud Architecture, Prompt Engineering, Reinforcement Learning.

Personal Interests: Content Creation: Youtube, Chess Strategy, Cross-cultural Communication, Language Learning, Creative Writing, Journalism, Dance, Debating, Stand Up Comedy, Public Speaking.